

“THERE IS AN **INCREDIBLE AMOUNT OF MATTER** BETWEEN US AND THE CENTER OF THE MILKY WAY TO **OBSCURE OUR VIEW.**”

Terry Oswalt, National Science Foundation program manager for Stellar Astronomy and Astrophysics, 1998

This color-enhanced X-ray image shows the region around Sagittarius A*, the supermassive black hole at the galactic center that produces regular X-ray flares as a result of vaporizing asteroids and other matter.

INTERNET USE BEGAN TO GROW steeply this year. A new, **high speed internet connection** through telephone lines became available in 1998. With asymmetric digital subscriber line (ADSL), users could receive information from the internet at 8 megabits (8 million bits) per second. The previous year, another **broadband technology** had its debut for home users: the cable modem, which connected to the internet via existing coaxial cables that also delivered television signals. These new technologies allowed users to easily **download larger files**, such as mp3 music files, more easily. The world's first **portable mp3 player**, the MPMan F10, was released this year, by SaeHan Information Systems from South Korean.

In September, American astronomer **Andrea Ghez** (b.1965) reported that she had detected the presence of a supermassive **black hole at the center of our galaxy**. Astronomers have since found evidence that supermassive black holes are present in most, if not all, galactic centers.

Astrophysicists studying supernovas in distant galaxies came to the conclusion that the **expansion of the Universe is accelerating**; this was the first concrete evidence of the existence of a cosmological constant—



Robotic surgery

The da Vinci Surgical System has four robot arms, which are controlled by the surgeon. One of the arms carries a high definition, 3-D vision system.

some kind of repulsive agent, also known as **dark energy** (see pp.344–45), that drives the expansion of space-time ever faster. Dark energy probably accounts for around three-quarters of the mass-energy in the Universe.

In November, a Russian Proton rocket launched the first module of the **International Space**

Station (ISS), into orbit. The ISS has been inhabited continuously since November 2000. The project involves **five space agencies** representing a total of 16 countries.

A team led by American cell biologist **James Thomson** (b.1958) created a culture of **human embryonic stem cells** (hESCs). These cells can develop into any kind of tissue, and have the potential of creating donor-matched organs for transplantation. The cultures, or lines, began with cells

harvested from human embryos—a fact that raised significant ethical concerns, since the technique led to the destruction of human embryos.



Worm genome
Caenorhabditis elegans was the first multicellular organism to have its genome sequenced. This nematode worm lives in soil and grows to about 1/32 in (1 mm) in length.

Thomson later managed to create induced stem cells—similar to hESCs, but created from **reprogrammed adult cells**, not taken from embryos (see 2007).

In another milestone in genomics (see 1977 and 1995), a nematode worm became the **first multicellular organism** to have its **genome sequenced**. Within two years, the draft sequence of the human genome was complete (see 2000).

In May, German surgeon **Friedrich-Wilhelm Mohr** (b.1951) performed the first **robotically assisted heart surgery**. In robotic surgery, the surgeon benefits from the assistance of computer-controlled instruments, which do not suffer from vibration or fatigue. With information and images relayed via the internet, an expert surgeon can carry out **operations in distant locations** using surgical robots remotely.

March Release of the first consumer mp3 player, the MPMan F10

April First experimental demonstration of a quantum computer

May First robotically assisted heart bypass

September Astrophysicists announce that the **expansion of the Universe is accelerating**

September 8 Astronomers conclude that there is a supermassive black hole at the center of our galaxy

October ADSL (asymmetric digital subscriber line) technology introduced for the first time in the USA

20 November First component of the **International Space Station** launched into orbit

November 6 James Thomson creates human embryonic stem cell lines (cultures)

December 11 First genome sequencing for a multicellular organism reported